

Exploratory Data Analysis of Student Performance Factors

AI-ML Minor Project 2

Submitted By

Ansh Arora

1. Introduction

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Student Performance Factors dataset and identify the factors that influence student academic performance. The analysis focuses on data cleaning, outlier detection, feature engineering, correlation analysis, and categorical data encoding to uncover meaningful insights.

Dataset Source

The dataset was obtained from Kaggle and contains information about students' academic habits, attendance, family background, learning environment, and examination performance.

Problem Statement

Educational institutions seek to understand the factors that contribute to student success. This project analyzes various student-related attributes to determine their relationship with examination scores and identify the most influential factors affecting academic performance.

2. Dataset Description

The dataset contains 6,607 student records and 20 features, including both numerical and categorical variables.

Data Dictionary

Feature	Description
Hours_Studied	Number of hours spent studying
Attendance	Student attendance percentage
Parental_Involvement	Level of parental involvement
Access_to_Resources	Availability of educational resources
Extracurricular_Activities	Participation in extracurricular activities
Sleep_Hours	Average sleep hours
Previous_Scores	Previous academic performance
Motivation_Level	Student motivation level
Internet_Access	Availability of internet access
Tutoring_Sessions	Number of tutoring sessions attended
Family_Income	Family income category
Teacher_Quality	Quality of teaching
School_Type	Type of school
Peer_Influence	Influence of peers
Physical_Activity	Time spent on physical activities
Learning_Disabilities	Presence of learning disabilities
Parental_Education_Level	Parents' education level
Distance_from_Home	Distance between home and school
Gender	Student gender
Exam_Score	Final examination score

3. Business Questions

1. Does study time affect student performance?
 2. Does attendance influence examination scores?
 3. Which variables are most strongly associated with exam scores?
 4. Are there any significant outliers in student behavior and performance?
 5. How do family and educational factors influence academic success?
-

4. Data Cleaning and Integrity Checks

Missing Value Analysis

The dataset was examined for missing values. Missing values were identified in the following categorical variables:

- Teacher_Quality
- Parental_Education_Level
- Distance_from_Home

Since these variables are categorical in nature, missing values were imputed using the mode (most frequent category). This method preserves the dataset size while maintaining representative values.

Duplicate Records

The dataset was checked for duplicate records using the duplicated() function. Any duplicate observations were removed to ensure data integrity and prevent biased analysis.

Data Type Verification

Data types were inspected using the info() function. Numerical variables were correctly stored as integer values, while categorical variables were stored as object/string data types. No major data type corrections were required.

5. Outlier Analysis

Skewness Analysis

Skewness was calculated for all numerical variables to understand their distributions. Most variables showed acceptable levels of skewness, indicating relatively balanced distributions.

Boxplot Analysis

Boxplots were generated for:

- Hours_Studied
- Attendance
- Sleep_Hours
- Previous_Scores
- Tutoring_Sessions
- Physical_Activity
- Exam_Score

These visualizations helped identify extreme observations and potential outliers.

IQR-Based Outlier Detection

The Interquartile Range (IQR) method was applied to detect outliers across numerical variables.

Key Findings

- A small number of students exhibited unusually high or low study hours.
- Attendance contained several extreme observations.
- Exam scores showed a limited number of exceptional performers.
- Most outliers represented genuine student behavior rather than data entry errors.

Outlier Handling

Outliers were retained in the dataset because they represent realistic student characteristics and academic performance levels. Removing these observations would result in loss of valuable information regarding exceptional students.

6. Bivariate Analysis

Hours Studied vs Exam Score

A scatter plot was created to examine the relationship between study hours and examination performance.

Observation

Students who spent more hours studying generally achieved higher examination scores, indicating a positive relationship between study effort and academic success.

Attendance vs Exam Score

A scatter plot was used to analyze attendance and examination performance.

Observation

Students with higher attendance percentages tended to perform better academically than students with lower attendance.

Parental Involvement vs Exam Score

A boxplot was used to compare exam scores across different parental involvement levels.

Observation

Students receiving higher parental involvement generally demonstrated better academic outcomes.

Internet Access vs Exam Score

A bar chart was used to compare examination performance based on internet availability.

Observation

Students with internet access achieved slightly higher average scores due to improved access to educational resources.

Family Income vs Exam Score

A bar chart was generated to evaluate the relationship between family income and examination performance.

Observation

Students from higher-income households generally performed better academically compared to students from lower-income families.

7. Multivariate Analysis

Correlation Heatmap

A correlation heatmap was constructed using all numerical variables to identify strong positive and negative relationships.

Strongest Positive Correlations with Exam Score

Variable	Correlation
Study_Attendance_Index	0.65
Attendance	0.58
Hours_Studied	0.44
Previous_Scores	0.18
Tutoring_Sessions	0.16

Weak Relationships

Variable	Correlation
Physical_Activity	0.03
Sleep_Hours	-0.02

Interpretation

Attendance and study effort were found to be the strongest factors associated with examination performance.

It is important to note that correlation does not imply causation. While these variables show statistical associations with exam scores, other hidden factors may also contribute to student performance.

8. Feature Engineering

A new feature named Study_Attendance_Index was created using the following formula:

$\text{Study_Attendance_Index} = \text{Hours_Studied} \times \text{Attendance}$

This feature combines student study effort and attendance into a single metric representing overall academic engagement.

Result

The engineered feature showed the highest correlation with Exam_Score (0.65), indicating that consistent attendance combined with dedicated study habits is strongly associated with academic success.

9. Encoding Preparation

Ordinal Variables

The following variables were treated as ordinal because their categories have a natural ranking:

- Parental_Involvement
- Access_to_Resources
- Motivation_Level
- Family_Income
- Teacher_Quality

Label Encoding was applied to these variables.

Nominal Variables

The following variables were treated as nominal because they do not possess a meaningful ranking:

- Extracurricular_Activities
- Internet_Access
- School_Type
- Peer_Influence
- Learning_Disabilities
- Gender

One-Hot Encoding was applied to these variables.

10. Key Findings

1. Attendance demonstrated a strong positive relationship with examination performance.
 2. Study hours significantly influenced student scores.
 3. The engineered Study_Attendance_Index emerged as the strongest predictor of academic performance.
 4. Previous academic achievement moderately influenced current examination scores.
 5. Internet access positively impacted student outcomes.
 6. Students with higher parental involvement generally performed better.
 7. Family income showed a positive association with academic performance.
 8. Sleep hours exhibited a weak relationship with examination scores.
 9. Most detected outliers represented genuine student behavior and were retained.
 10. Academic engagement appears to be the most important factor contributing to student success.
-

11. Conclusion

This project successfully applied Exploratory Data Analysis techniques to investigate factors influencing student academic performance.

The analysis revealed that attendance and study effort are the most important variables associated with examination success. The newly created Study_Attendance_Index displayed the strongest relationship with Exam_Score, highlighting the importance of consistent participation and dedicated study habits.

Although factors such as previous scores, family income, parental involvement, and internet access also contributed to student performance, their influence was lower compared to attendance and study hours.

The dataset was successfully cleaned, analyzed, and prepared for future predictive modeling through feature engineering and categorical encoding techniques.

Overall, the findings suggest that improving attendance and encouraging effective study habits can significantly enhance academic outcomes.

12. References

1. Kaggle Student Performance Factors Dataset
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation